

Topological Manifolds

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Abstract

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1 Structural Theorems of Topological Manifolds

1.1 The Topology of Manifolds

Theorem 1.1.1 (Invariance of Dimension)

Let M be a topological manifold of dimension n and m . Then $n = m$.

Theorem 1.1.2 (Whitney Embedding Theorem for Topological Manifolds)

Let M be a compact topological manifold. Then there exists an embedding $M \hookrightarrow \mathbb{R}^{2 \dim(M)+1}$.

1.2 CW Complex Structures

Proposition 1.2.1

Let M be a topological manifold that is a CW complex. Then the dimension of M as a manifold is the dimension of M as a CW complex.

Proposition 1.2.2

Let M be a compact topological manifold. Then M is homotopy equivalent to a finite CW complex.

1.3 Covering Spaces of Manifolds

Proposition 1.3.1 Let M be a manifold. Let $p : \widetilde{M} \rightarrow M$ be a covering space. Then $\widetilde{M}$ is also a manifold.

2 Algebraic Topology of Topological Manifolds

2.1 Fundamental Results

Lemma 2.1.1 Let M be a connected k -dimensional manifold. Let R be a ring. Then we have

$$H_i(M; R) = 0$$

for all $i \geq k$.

The remainder of the subsection concerns compact manifolds.

Proposition 2.1.2 Let M be a compact manifold. Then $H_k(M)$ is finitely generated for all $k \in \mathbb{N}$.

Proposition 2.1.3 Let M be a compact manifold. If $\dim(M)$ is odd, then $\chi(M) = 0$.

2.2 Local and Global Orientations

The key observation in defining orientation through homology is the following proposition, which shows that the local homology groups on a manifold are isomorphic to R on the top dimension.

Proposition 2.2.1

Let M be a k -dimensional topological manifold and $x \in M$ a point. Let R be a ring. Then

$$H_n(M, M \setminus \{x\}; R) \cong \begin{cases} R & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

Proof Let $x \in M$. Since M is a manifold, there exists an open neighbourhood of x such that $U \cong \mathbb{R}^k$ via some transition map $\varphi : U \rightarrow \mathbb{R}^k$. Since $M \setminus U$ is closed, we can apply excision to obtain an isomorphism

$$H_n(M \setminus \{x\}; R) \cong H_n(U \setminus \{x\}; R)$$

By the homeomorphism $(U, U \setminus \{x\}) \cong (\mathbb{R}^k, \mathbb{R}^k \setminus \{\varphi(x)\})$ and 6.3.2 in AT2, we obtain the desired result. $\blacksquare$

Definition 2.2.2 (The Orientation Cover)

Let M be a topological manifold of dimension n . Let R be a ring. Define the orientation bundle of M to be the set

$$\widetilde{M}_R = \{(p, \omega_p) \mid p \in M \text{ and } \omega_p \text{ is a generator of } H_n(M, M \setminus \{p\}; R)\}$$

together with topology given as follows. Let $\pi : \widetilde{M} \rightarrow M$ be the projection function. Let $\{(U_i, \phi_i) \mid i \in I\}$ be an atlas of M . Let

$$\alpha_{U_i, p} : H_n(M, M \setminus U_i; R) \rightarrow H_n(M, M \setminus \{p\}; R)$$

be the map induced by inclusion. Then the topology of $\widetilde{M}$ is generated by the sets $\{(p, \alpha_{U_i, p}(\omega_p)) \mid p \in U\}$.

Definition 2.2.3 (R-Orientation)

Let M be a topological manifold. Let R be a ring. An R -orientation of M is a section of the orientation cover $\widetilde{M}_R \rightarrow M$.

Proposition 2.2.4

Let M be a topological manifold. Let R be a ring. Then the following are true.

- Suppose that R has n generators. Then $\widetilde{M}$ is an n -sheeted covering space of M .

- $\widetilde{M}$ is an orientable manifold.

Proposition 2.2.5

Let M be a connected topological manifold. Then the following are true.

- M is $\mathbb{Z}$ -orientable if and only if $\widetilde{M}_{\mathbb{Z}} \cong M \amalg M$.
- M is not $\mathbb{Z}$ -orientable if and only if $\widetilde{M}_{\mathbb{Z}}$ is a non-trivial 2-sheeted cover.

Proposition 2.2.6

Let M be a topological manifold. Let R be a ring. Then the following are true.

- If M is $\mathbb{Z}$ -orientable, then M is R -orientable.
- If M is not $\mathbb{Z}$ -orientable, then M is R -orientable if and only if R contains a unit of order 2.

Definition 2.2.7 (Set of Sections) Let M be a compact k -manifold. Denote $p : M_R \rightarrow M$ the projection map. Define the set of sections of M to the orientation cover to be the set

$$\Gamma(M, M_R) = \{s : M \rightarrow M_R \mid s \circ p = \text{id}_M\}$$

Proposition 2.2.8 Let M be a k -manifold. Let $K \subseteq M$ be compact. Then the following are true.

- If $s : M \rightarrow M_R$ is a section sending x to (x, μ_x) , there exists a unique $\mu_K \in H_k(M \mid K; R)$ such that under induced map of inclusions

$$H_k(M \mid K; R) \rightarrow H_k(M \mid x; R)$$

μ_K is mapped to μ_x .

- The local homology groups $H_i(M \mid K; R) = 0$ for all $i > k$.

Proof Step 1: If the statement is true for compact subsets $A, B, A \cap B$, then it is true for $A \cup B$. We obtain an exact sequence:

$$0 \longrightarrow H_k(M \mid A \cup B; R) \xrightarrow{\Phi} H_k(M \mid A; R) \oplus H_k(M \mid B; R) \xrightarrow{\Psi} H_k(M \mid A \cap B; R)$$

using the Mayer Vietoris sequence. Since $H_i(M \mid A; R)$ and $H_i(M \mid B; R)$ is 0 for all $i > k$, we conclude that $H_i(M \mid A \cup B; R) = 0$ for all $i > k$. This proves the second half of the proposition. For the first half, let $s : M \rightarrow M_R$ be a section sending x to (x, μ_x) . Then we obtain unique classes $\mu_A \in H_k(M \mid A; R)$ and $\mu_B \in H_k(M \mid B; R)$ and $\mu_{A \cap B} \in H_k(M \mid A \cap B; R)$ such that for $x \in A \cap B$, μ_A, μ_B and $\mu_{A \cap B}$ are all mapped to μ_x in the maps

$$H_k(M \mid A; R), H_k(M \mid B; R), H_k(M \mid A \cap B; R) \rightarrow H_k(M \mid x; R)$$

Then $\Psi(\mu_A, -\mu_B) = \mu_{A \cap B} - \mu_{A \cap B} = 0$ implies that $(\mu_A, -\mu_B) \in \ker(\Psi)$. By exactness of the above sequence, there exists $\mu_{A \cup B} \in H_k$ such that $\Phi(\mu_{A \cup B}) = (\mu_A, -\mu_B)$.

For uniqueness, suppose that $\mu_{A \cup B}$ and $\mu'_{A \cup B}$ be two such elements satisfying the requirements. Consider the following diagram:

$$\begin{array}{ccccc} 0 & \longrightarrow & H_k(M \mid A \cup B; R) & \xrightarrow{\Phi} & H_k(M \mid A; R) \oplus H_k(M \mid B; R) & \xrightarrow{\Psi} & H_k(M \mid A \cap B; R) \\ & & & & \downarrow & & \\ & & & & H_k(M \mid x; R) & & \end{array}$$

It is commutative since Φ and Ψ are induced by inclusions. Then the image of $\mu_{A \cap B} - \mu'_{A \cap B}$ is 0 in $H_k(M \mid x; R)$. By commutativity of the left triangle, the image of $\mu_A - \mu'_A$ and $\mu_B - \mu'_B$ is 0 in $H_k(M \mid x; R)$. Since Φ is injective, this implies that $\mu_{A \cap B} = \mu'_{A \cap B}$.

Step 2: Reduce to the case of $M = \mathbb{R}^n$.

If $K \subseteq M$ is compact, then there exists compact subsets $K_1, \dots, K_n$ of M , each contained in a local chart and hence homeomorphic to a subset of $\mathbb{R}^n$, such that $K = K_1 \cup \dots \cup K_n$. By step 1, it

suffices to consider K_1 . Then by excision, it suffices to consider a compact subset K of $\mathbb{R}^n$.

Step 3: Reduce to the case of a convex compact subset of $\mathbb{R}^n$. ■

Proposition 2.2.9 Let M be a compact k -manifold. Then the following are true.

- If M is R -orientable, then there is an R -module isomorphism

$$\Gamma(M, M_R) \cong \text{Hom}(\pi_0(M), R)$$

of sets. This assignment is given by sending $s : \pi_0(M) \rightarrow R$ to the map $x \mapsto (x, s([x]))$.

- If M is connected and not R -orientable, then there are no global sections.

Theorem 2.2.10 Let M be a compact and connected k -dimensional manifold. Let R be a ring. Then the following are true.

- If M is R -orientable, then the map

$$H_k(M; R) \rightarrow H_k(M, M \setminus \{x\}; R) \cong R$$

is an isomorphism for all $x \in M$.

- If M is not R -orientable, then the map

$$H_k(M; R) \rightarrow H_k(M, M \setminus \{x\}; R) \cong R$$

is injective and has image $\{r \in R \mid 2r = 0\}$ for all $x \in M$

Notice this theorem is not a definitive criterion for orientability in general. However, if $R = \mathbb{Z}$, then it becomes a sufficient criterion.

We can summarize the situation as follows:

$$M \text{ is orientable} \implies \begin{array}{c} \text{TF AE:} \\ M \text{ is } R\text{-orientable} \\ \text{there exists a section } M \rightarrow M_R \end{array} \implies \begin{array}{c} \Gamma(M, M_R) \cong \text{Hom}(\pi_0(M), R) \\ H_k(M; R) \cong H_k(M \setminus \{x\}; R) \cong R \end{array}$$

$$\begin{array}{c} \text{TF AE:} \\ M \text{ is not } R\text{-orientable} \\ \text{there are no sections } M \rightarrow M_R \end{array} \implies H_k(M; R) \longrightarrow H_k(M \setminus \{x\}; R) \cong R \text{ is injective}$$

Corollary 2.2.11 Let M be a compact and connected k -dimensional manifold. Then the following are true.

- If M is orientable, then the torsion part of $H_{k-1}(M)$ is trivial.
- If M is not-orientable, then the torsion part of $H_{k-1}(M)$ is $\mathbb{Z}/2\mathbb{Z}$.

Corollary 2.2.12 Any simply connected manifold is orientable.

Proof By Galois theory of covering spaces, any 2-sheeted cover of a simply connected space is disconnected. ■

Proposition 2.2.13 Let $k \geq 1$. Then $\mathbb{RP}^k$ is orientable if and only if k is odd.

Proof The quotient map $q : S^k \rightarrow \mathbb{RP}^k$ is the unique connected two-sheeted cover of $\mathbb{RP}^k$ by Galois theory for covering spaces. The non-trivial deck transformation is given by the antipodal map which has degree $(-1)^{k+1}$. If k is odd then this degree is 1 so that the deck transformation is orientation preserving. Since deck transformations of the orientation bundle must be orientation reversing, we conclude that $S^k \neq \widetilde{\mathbb{RP}^k}$. This means that the orientation bundle of $\mathbb{RP}^k$ is disconnected.

Now assume that k is even. By the lifting criterion, there exists a lift of q called $\tilde{q}$ such that

$$\begin{array}{ccc} & \widetilde{\mathbb{RP}^k} & \\ \tilde{q} \nearrow & \downarrow p & \\ S^k & \xrightarrow{q} & \mathbb{RP}^k \end{array}$$

where p is the covering map. Then $\tilde{q}$ must also be a covering space. Assume that q is not injective. This means that $\tilde{q} \circ (-\text{id}) = \tilde{q}$ since $-\text{id}$ is the only other deck transformation of S^k over $\mathbb{RP}^k$. This means that for any $x \in S^k$, we have that

$$H_k(S^k) \xrightarrow{\tilde{q}} H_k(\widetilde{\mathbb{RP}^k}) \longrightarrow H_k(\widetilde{\mathbb{RP}^k}, \widetilde{\mathbb{RP}^k} \setminus \{\tilde{q}(x)\})$$

where the second map is given by the long exact sequence in relative homology. Denoting this entire map by α , we have that $\alpha \circ (-\text{id})_* = \alpha$ since $\tilde{q} \circ (-\text{id}) = \tilde{q}$. But α is a map from $\mathbb{Z}$ to $\mathbb{Z}$. Since $\alpha \circ (-\text{id})_* = \alpha$ this implies that $\alpha = 0$. But α also factors as

$$H_k(S^k) \xrightarrow{\cong} H_k(S^k, S^k \setminus \{x\}) \xrightarrow{\tilde{q}} H_k(\widetilde{\mathbb{RP}^k}, \widetilde{\mathbb{RP}^k} \setminus \{\tilde{q}(x)\})$$

by the long exact sequence in relative homology and naturality. But the second map is also an isomorphism since covering spaces of manifolds induces a an isomorphism in local homology groups.

Now S^k being compact and $\mathbb{RP}^k$ being Hausdorff together with $\tilde{q}$ being injective implies that $\tilde{q}$ is a homeomorphism onto an open and closed subspace of $\mathbb{RP}^k$. Assume that $\tilde{q}$ is not surjective, then we have that $\widetilde{\mathbb{RP}^k} \cong S^k \amalg X$ for some other space X . But this is impossible thus q is surjective and $\tilde{q}$ gives a homeomorphism between S^k and $\mathbb{RP}^k$. Since S^k is connected, $\mathbb{RP}^k$ is thus non orientable. ■

One has to be careful that homotopy equivalence does not preserve orientability. For example, the Möbius strip is homotopy equivalent to S^1 but the former is non-orientable while the latter is.

2.3 Fundamental Class

Definition 2.3.1 (Fundamental Class) Let M be a compact and connected manifold of dimension n . Let R be a ring. A fundamental class for M with coefficients in R is an element $[M] \in H_n(M; R)$ such that the element $[M]$ is sent to a generator under the induced map of inclusion

$$H_n(M; R) \rightarrow H_n(M \mid x; R) \cong R$$

for any $x \in M$.

Lemma 2.3.2 Let M be a compact and connected manifold of dimension n . Let R be a ring. Then M is R -orientable if and only if M has a fundamental class for M with coefficients in R .

When M is a δ -complex, we can represent the fundamental class as a linear combination of top-simplices satisfying some conditions.

Proposition 2.3.3 Let M be a compact and connected n -manifold. Let

$$\rho = \sum_{\sigma_i \text{ is an } n \text{ simplex}} k_i \sigma_i \in C_n(M)$$

be an n -chain. Then $[\rho]$ is a fundamental class of M if and only if each $k_i = \pm 1$ and ρ is an n -cycle. Moreover, M is orientable if and only if there exists such a ρ .

We can explicitly give a fundamental class of the sphere in integral coefficients.

Proposition 2.3.4 Let $\sigma : \Delta^{k+1} \rightarrow \Delta^{k+1}$ be the identity singular n -simplex in the space Δ^{k+1} . Then the cycle $\partial\sigma \in C_k(\partial\Delta^{k+1})$ represents a generator in for the top homology of S^k .

Proof It is clear that it is a cycle since it is a boundary in the chain complex $C_\bullet(\Delta^{k+1})$. We proceed by induction. When $k = 0$, the statement is clear. So suppose that $k > 0$. Let U_1, U_2 be open subspaces of Δ^{k+1} as follows. U_1 is an open neighbourhood of the last face of $\partial_{k+1}\Delta^{k+1}$ which deformation retracts onto $\partial\Delta^{k+1}$. U_2 is an open neighbourhood of the remaining faces $U_2 = \bigcup_{i=0}^k \partial_i\Delta^{k+1}$ which deformation retracts onto $\bigcup_{i=0}^k \partial_i\Delta^{k+1}$. Moreover, choose them in such a way that $U_1 \cap U_2$ deformation retract onto $\partial\Delta^{k+1} = \partial[v_0, \dots, v_k]$ and $U_1 \cup U_2$ deformation retracts onto $\partial[v_0, \dots, v_{k+1}]$. By induction hypothesis, we know that $\tilde{H}_{k-1}(U_1 \cap U_2) \cong \mathbb{Z}$ is generated by

$$\partial([v_0, \dots, v_k]) = \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k]$$

From Mayer-Vietoris sequence, since $U_1 \cup U_2$ deformation retracts onto $\partial\Delta^{k+1}$, the connecting homomorphism

$$\tilde{H}_k(U_1 \cup U_2) \rightarrow \tilde{H}_{k-1}(U_1 \cap U_2)$$

since U_1 and U_2 are contractible so we only need to show that $\partial\sigma$ is sent to the generator or its negative.

For this we will explicitly compute the connecting homomorphism. It is clear that

$$\left((-1)^{k+1} [v_0, \dots, v_k], \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_{k+1}] \right) \in C_k(U_1) \oplus C_k(U_2)$$

is such that it is a lift of the cycle $\partial\sigma$. Its image under the connecting homomorphism is the unique $(k-1)$ -cycle τ in $U_1 \cap U_2$ which satisfies

$$((i_1)_*(\tau), -(i_2)_*(\tau)) = \left((-1)^{k+1} \partial([v_0, \dots, v_k]), \sum_{i=0}^k (-1)^i \partial([v_0, \dots, \hat{v}_i, \dots, v_{k+1}]) \right)$$

It is clear that $\tau = (-1)^{k+1} \partial([v_0, \dots, v_k])$ is a generator in $\tilde{H}_k(U_1 \cap U_2)$. ■

Corollary 2.3.5 Let S_+^k and S_-^k be the northern and southern hemisphere of S^k respectively. Choose homomorphisms

$$\sigma_+ : \Delta^k \xrightarrow{\cong} S_+^k \quad \text{and} \quad \sigma_- : \Delta^k \xrightarrow{\cong} S_-^k$$

such that both σ_+, σ_- map the boundary $\partial\Delta^k$ homeomorphically onto the equator $S_+^k \cap S_-^k$ and the composition

$$\partial\Delta^k \xrightarrow{\sigma_+} S_+^k \cap S_-^k \xrightarrow{(\sigma_-)^{-1}} \partial\Delta^k$$

is the identity. Then the cycle $\sigma_+ - \sigma_- \in C_k(S^k)$ represents a fundamental class for S^k .

Proof For $k = 1$, $\sigma_+ : \Delta^1 \rightarrow S^1$ is the upper half circle oriented anticlockwise and $\sigma_- : \Delta^1 \rightarrow S^1$ is the lower half circle oriented clockwise. It is clear that by the isomorphism $\pi_1(S^1, 1)^{\text{ab}} \cong H_1(S^1)$, $\sigma_+ - \sigma_-$ is a generator. Now assume that $k > 1$. It is clear from the assumptions that $\sigma_+ - \sigma_-$ is a cycle. Choose open neighbourhoods U_+ and U_- of S_+^k and S_-^k respectively which deformation retracts onto S_+^k and S_-^k and that $U_+ \cap U_- \simeq S^{k-1}$ the equator. The connecting homomorphism

$$H_k(S^k) \rightarrow H_{k-1}(U_+ \cap U_-)$$

in the Mayer-Vietoris sequence is an isomorphism that sends $\sigma_+ - \sigma_-$ to $\partial\sigma_+ = \partial\sigma_-$. By the above proposition, $\partial\sigma_+ = \partial\sigma_-$ is a generator of $H_{k-1}(\partial\Delta^k)$ and so we are done. ■

2.4 The Cap Product

Definition 2.4.1 (The Cap Product) Let X be a space. Let R be a commutative ring. Let $\sigma|_{[v_0, \dots, v_k]} : \Delta^k \rightarrow X$ be a singular k -simplex and $\phi \in C^l(X; R)$. Define the cap product to be

$$\sigma \frown \phi = \phi(\sigma|_{[v_0, \dots, v_l]})\sigma|_{[v_l, \dots, v_k]} \in C_{k-l}(X)$$

Lemma 2.4.2 Let X be a space. Let R be a commutative ring. Let $\sigma|_{[v_0, \dots, v_k]} : \Sigma^k \rightarrow X$ be a k -simplex and $\phi \in C^l(X; R)$. Then we have

$$\partial(\sigma \frown \phi) = (-1)^l(\partial(\sigma) \frown \phi - \sigma \frown \delta(\phi))$$

Proposition 2.4.3 Let X be a space. Let R be a commutative ring. Then the cap product descends to homology

$$\frown : H_k(X; R) \times H^l(X; R) \rightarrow H_{k-l}(X; R)$$

and is well defined for $k \geq l$.

Lemma 2.4.4 Let X, Y be spaces. Let R be a commutative ring. Let $f : X \rightarrow Y$ be a map. Let $\alpha \in H_k(X; R)$ and $\phi \in H^l(Y; R)$. Then we have

$$f_*(\alpha) \frown \phi = f_*(\alpha \frown f^*(\phi))$$

We can describe this lemma somewhat as a naturality condition:

$$\begin{array}{ccccc} H_k(X; R) \times H^l(X; R) & \xrightarrow{\frown} & H_{k-l}(X; R) \\ f_* \downarrow & \uparrow f^* & \downarrow f_* \\ H_k(Y; R) \times H^l(Y; R) & \xrightarrow{\frown} & H_{k-l}(Y; R) \end{array}$$

Lemma 2.4.5 Let X be a space. Let R be a commutative ring. Let $\alpha \in C_{k+l}(X; R)$ and $\phi \in C^k(X; R)$ and $\psi \in C^l(X; R)$. Then we have

$$\psi(\alpha \frown \phi) = (\phi \smile \psi)(\alpha)$$

Proposition 2.4.6 Let X be a space. Let R be a commutative ring. Let $\phi \in C^k(X; R)$ and $\psi \in C^l(X; R)$. Then the following diagram commutes:

$$\begin{array}{ccc} H^k(X; R) & \xrightarrow{h} & \text{Hom}_{\mathbb{Z}}(H_k(X), R) \\ \phi \smile - \downarrow & & \downarrow (-\frown \psi)^* \\ H^{k+l}(X; R) & \xrightarrow{h} & \text{Hom}_{\mathbb{Z}}(H_{k+l}(X), R) \end{array}$$

In particular, if the maps h are isomorphisms, then $-\frown \psi$ is the dual map of $\phi \smile -$.

2.5 Poincare Duality

Proposition 2.5.1 Let R be a commutative ring. Let M be a connected R -orientable manifold of dimension n . Then the duality map

$$D : H_c^p(M; R) \rightarrow H_{n-p}(M)$$

defined by $D(\alpha) = \alpha \frown [M]$ is an isomorphism.

Theorem 2.5.2 (Poincare Duality) Let R be a commutative ring. Let M be a compact connected

R -orientable manifold of dimension n . Then the map

$$D : H^p(M; R) \rightarrow H_{n-p}(M; R)$$

defined by $D(\alpha) = \alpha \frown [M]$ is an isomorphism.

Corollary 2.5.3 Let M be a compact connected $\mathbb{F}$ -orientable manifold of dimension n . Let $\mathbb{F}$ be a field. Then the Poincare duality induces isomorphisms

$$H^k(M; \mathbb{F}) \cong H_{n-k}(M; \mathbb{F})$$

Proposition 2.5.4 Let M be a compact connected orientable manifold of dimension n . Let R be a commutative ring. Suppose that $H^k(X; R)$ is a free R -module for all k . Then the bilinear map

$$H^k(M; R) \times H^{n-k}(M; R) \rightarrow R$$

defined by $(\phi, \psi) \mapsto [M] \frown (\phi \smile \psi) = (\phi \smile \psi)([M])$ is a perfect pairing.

Given a compact connected $\mathbb{Z}$ -orientable topological manifold M of dimension n , we know the following.

- All homology groups and hence cohomology groups are finitely generated.
- $H_n(M) \cong \mathbb{Z}$ and $H_{n-1}(M)$ is torsion-free.
- $H^k(M; \mathbb{Z}) \cong H_{n-k}(M)$ for $0 \leq k \leq n$.
- The free part of cohomology satisfies the following:

$$H^k(M; R) \times H^{n-k}(M; R) \rightarrow R$$

given by $(\phi, \psi) \mapsto (\phi \smile \psi)([M])$ is a perfect pairing.

- In particular this means that the free part of homology and cohomology are palindromic.

If instead M is non-orientable, then we have the following.

- $H_n(M) = 0$ and the torsion part of $H_{n-1}(M)$ is $\mathbb{Z}/2\mathbb{Z}$.

2.6 Alexander and Lefschetz Duality

Theorem 2.6.1 (Alexander Duality) Let $n \in \mathbb{N}$. Let $K \subset S^n$ be compact, and such that it has a neighbourhood whose retract is K . Then there are isomorphisms

$$\tilde{H}_i(S^n \setminus K) \cong \tilde{H}^{n-i-1}(K)$$

3 Operations on Topological Manifolds

3.1 Connected Sums

Definition 3.1.1 (Connected Sum of Manifolds)

Let M_1, M_2 be topological manifolds of dimension n . Let $i_1 : D^n \rightarrow M_1$ and $i_2 : D^n \rightarrow M_2$ be embeddings. Define the connected sums of M_1 and M_2 via the discs $i_1(D^n)$ and $i_2(D^n)$ to be

$$M_1 \# M_2 = \frac{M_1 \setminus i_1(0) \amalg M_2 \setminus i_2(0)}{i_1(tu) \sim i_2((1-t)u)}$$

for $u \in S^{n-1} \subseteq D^n$ and $0 < t < 1$.

Lemma 3.1.2

Let M be a topological manifold of dimension n . Then $M \# S^n \cong M$.

Lemma 3.1.3

Let M_1, M_2 be connected topological manifolds of dimension n . Then $M_1 \# M_2$ is connected.

Proposition 3.1.4

Let M_1, M_2 be closed connected topological manifolds of dimension n . Then the following are true.

- We have

$$H_k(M_1 \# M_2; \mathbb{Z}) \cong H_k(M_1; \mathbb{Z}) \oplus H_k(M_2; \mathbb{Z})$$

for $0 < k < n - 1$.

- If at least one of M_1, M_2 is orientable, then we have

$$H_{n-1}(M_1 \# M_2; \mathbb{Z}) \cong H_{n-1}(M_1; \mathbb{Z}) \oplus H_{n-1}(M_2; \mathbb{Z})$$

- If both M_1, M_2 are non-orientable, then we have $H_{n-1}(M_1 \# M_2)$ is obtained from $H_{n-1}(M_1; \mathbb{Z}) \oplus H_{n-1}(M_2; \mathbb{Z})$ by replacing one of the two copies of $\mathbb{Z}/2\mathbb{Z}$ with $\mathbb{Z}$.
- We have

$$\chi(M_1 \# M_2) = \chi(M_1) + \chi(M_2) - \chi(S^n)$$

4 Manifolds of Dimension 1

4.1 Topological Properties

Proposition 4.1.1

Let M be a topological manifold of dimension 1. Then M admits a CW complex structure.

4.2 Classification Results

Proposition 4.2.1

Let M be a connected topological manifold of dimension 1. Then M is homeomorphic to one of the following.

- The real line $\mathbb{R}$.
- The circle S^1 .

Proposition 4.2.2

Let M be a connected topological manifold with non-empty boundary of dimension 1. Then M is homeomorphic to one of the following.

- The interval $[0, 1]$.
- The interval $[0, \infty)$.

5 Manifolds of Dimension 2

5.1 Polygonal Presentations

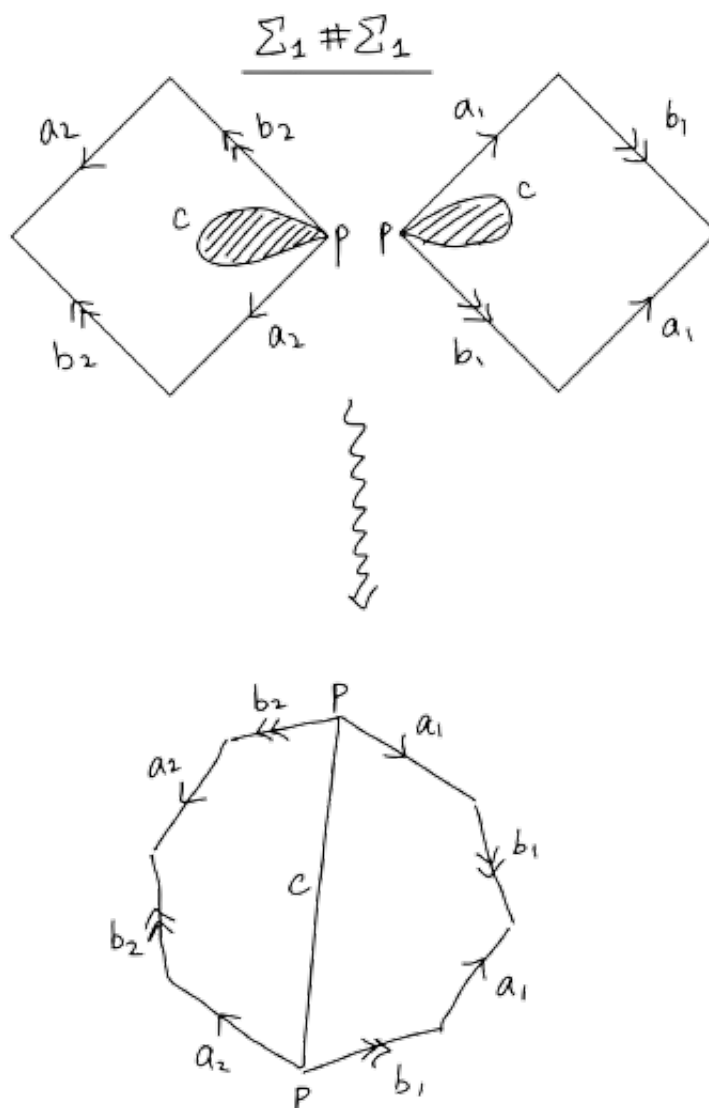
5.2 Classification of Compact Surfaces

Definition 5.2.1 (*g -Holed Torus*) For $g \geq 0$, define the g -holed torus to be

$$\Sigma_g = \mathbb{T} \# \cdots \# \mathbb{T}$$

the connected sum of g toruses. By convention when $g = 0$, Σ_g is the 2-sphere.

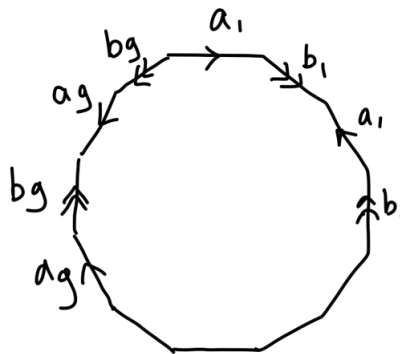
Recall the CW complex of the torus. We can visualize the connected sum of two toruses using the CW complex.



This is done by cutting a hole at the CW complex at the point p , and then pushing the boundary c out, and then connecting them together. The cut-out hole is exactly a disc in the torus. By gluing the two toruses along the boundary c , we are effectively gluing the two toruses along the discs.

The new heptagon obtained is precisely then the CW complex of Σ_2 . In general, we can perform the

operation of connected sum on a $(4g - 4)$ -gon and a square. We then obtain the CW complex of the g -holed torus.



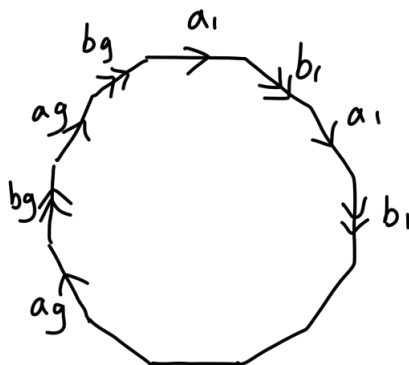
Another class of compact surfaces is the connected sum of projective spaces $\mathbb{RP}^2$.

Definition 5.2.2 (Non-Orientable Surface) For $h \geq 1$, define

$$N_h = \mathbb{RP}^2 \# \cdots \# \mathbb{RP}^2$$

the connected sum of h projective spaces.

We can do the same process of gluing the CW complexes just like that of the torus to obtain the $4h$ -gon that represents N_h :



It is also meaningful to ask what would happen if we perform connected sums through the two class of compact surfaces. We obtain the following.

Proposition 5.2.3 Let N_3 denote the connected sum of three projective spaces $\mathbb{RP}^2$. Then we have that

$$T \# \mathbb{RP}^2 = N_3$$

The above two classes of compact surfaces together with the sphere exhausts all possible cases for compact surfaces.

Theorem 5.2.4 Every compact surface is homeomorphic to exactly one of the following.

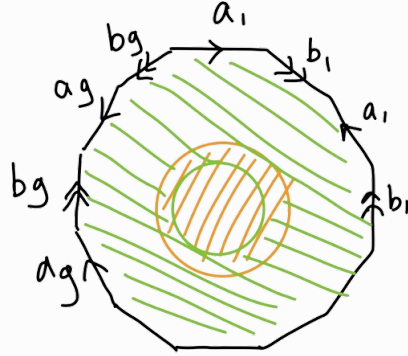
- Σ_g for $g \geq 0$
- N_h for $h \geq 1$

5.3 Homology and Cohomology of Compact Orientable Surfaces

Proposition 5.3.1 Let $g \geq 0$. The homology of the g -holed torus Σ_g is given by

$$H_n(\Sigma_g) = \begin{cases} \mathbb{Z} & \text{if } n = 0, 2 \\ \mathbb{Z}^{2g} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof Cut an open disc along the middle of the CW complex as follows



and label it V (the orange part). Label the green part as U . It is clear that $U \cap V \simeq S^1$, U is contractible and V deformation retracts to the boundary, which is actually just a wedge sum of $2g$ circles. By the formula for the homology of wedge sums we have that

$$H_n(V) = \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ \mathbb{Z}^{2g} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

By the reduced Mayer-Vietoris sequence, the only non-trivial homology groups in the sequence are

$$0 \longrightarrow \tilde{H}_2(\Sigma_g) \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^{2g} \longrightarrow \tilde{H}_1(\Sigma_g) \longrightarrow 0$$

and the exact sequence

$$0 \longrightarrow \tilde{H}_0(\Sigma_g) \longrightarrow 0$$

in which the latter immediately shows that $H_0(\Sigma_g) \cong \mathbb{Z}$. Now the map $\mathbb{Z} \rightarrow \mathbb{Z}^{2g}$ sends a generator of the first homology of $U \cap V \simeq S^1$ to the loop

$$a_1 + b_1 - a_1 - b_1 + \cdots + a_g + b_g - a_g - b_g$$

Since $\mathbb{Z}^{2g}$ is abelian, we conclude that this map is actually the zero map. It follows that $H_2(\Sigma_g) \cong \mathbb{Z}$ and $H_1(\Sigma_g) \cong \mathbb{Z}^{2g}$. ■

We can immediately deduce the orientability of Σ_g using the machinery in section 1.

Corollary 5.3.2 The surfaces Σ_g for $g \geq 0$ is orientable.

Proof By the above, we have that $H_2(\Sigma_g) \cong \mathbb{Z}$. The long exact sequence for relative homology groups give

$$\cdots \longrightarrow H_2(\Sigma_g \setminus \{x\}) \longrightarrow H_2(\Sigma_g) \longrightarrow H_2(\Sigma_g, \Sigma_g \setminus \{x\}) \longrightarrow H_1(\Sigma_g \setminus \{x\}) \longrightarrow H_1(\Sigma_g) \longrightarrow \cdots$$

Let U be as the proof above. Then the inclusion from U to $\Sigma \setminus \{x\}$ is a homotopy equivalence. Moreover, $\Sigma \setminus \{x\}$ is a $2g$ -fold wedge of circles labelled $a_1, b_1, \dots, a_g, b_g$ and $H_2(\Sigma_g \setminus \{x\}) = 0$. Also, we have that $H_1(U) \cong H_1(\Sigma_g)$ from above and hence $H_1(\Sigma_g \setminus \{x\}) \cong H_1(\Sigma_g)$. The last map is invertible so that by exactness, the third map is the zero map. Then what remains is an

isomorphism

$$H_2(\Sigma_g) \cong H_2(\Sigma_g, \Sigma_g \setminus \{x\})$$

Now since this isomorphism factors through $H_2(\Sigma_g, \Sigma_g \setminus B)$ for any ball B containing x , we thus have a consistent local orientation throughout all of Σ_g . ■

Proposition 5.3.3 Let $g \geq 0$. Let A be an abelian group. The singular cohomology of the g -holed torus Σ_g with coefficients in A is given by

$$H^n(\Sigma_g; A) = \begin{cases} A & \text{if } n = 0, 2 \\ A^{2g} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof Applying the universal coefficient theorem easily gives all the required cohomology groups. ■

We use the cohomology of Σ_g to illustrate generators of cohomology. Ref: Hatcher Ex3.7.

Proposition 5.3.4 Let $g \geq 0$. The integral cohomology ring of Σ_g is given by

$$H^*(\Sigma_g; \mathbb{Z}) \cong \frac{\mathbb{Z}[\alpha_1, \beta_1, \dots, \alpha_g, \beta_g]}{(\alpha_i^2, \beta_i^2, \alpha_i \alpha_j, \beta_i \beta_j, \alpha_i \beta_j, \beta_i \alpha_j, \alpha_i \beta_i + \beta_i \alpha_i \mid 1 \leq i \neq j \leq g)} \cong \Lambda^2(\mathbb{Z}^{2g})$$

where each α_i and β_i are of degree 1 in the graded ring.

Proof Consider the following CW complex of Σ_g . Recall that a basis for $H_1(\Sigma_g)$ is given by $a_1, b_1, \dots, a_g, b_g$. Consider the dual basis $\alpha_1, \beta_1, \dots, \alpha_g, \beta_g$. By definition these are precisely the generators of $H^1(\Sigma_g; \mathbb{Z})$. By definition the value of α_i is 1 on a_i and 0 otherwise. This is similar for β_i . We want to find elements of $Z^1(\Sigma_g; \mathbb{Z})$ that represent α_i and β_i . Consider the following diagram: Define $\phi_i \in C^1(\Sigma_g; \mathbb{Z})$ to be the map that gives 1 for any edge that intersects with s_i and 0 otherwise. Similarly, define $\psi_i \in C^1(\Sigma_g; \mathbb{Z})$ to be the map that gives 1 for any edge that intersects with t_i and 0 otherwise. It is easy to see that $\delta(\phi_i) = 0$ and $\delta(\psi_i) = 0$ so that ϕ and ψ are indeed cocycles. Moreover, they represent α_i and β_i respectively.

Now notice that I have indicated orientations for each 2-simplices in Σ_g depending on whether they are oriented clockwise or anti-clockwise. Define an element of $C_2(\Sigma_g)$ by the sum of the 2-simplices with ± 1 as their coefficient depending on their orientation. It is easy to see that the sum is a 2-cycle that generates $H_2(\Sigma_g)$. Let γ be its dual generator.

It is easy to check that

$$\phi_i \smile \psi_j = -\psi_j \smile \phi_i = \begin{cases} \gamma & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

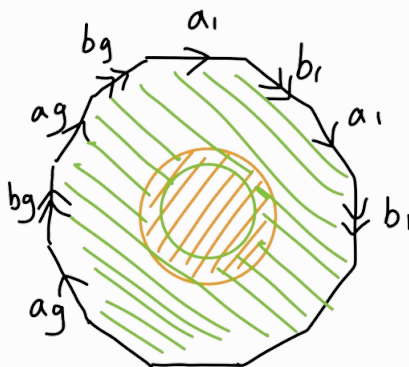
and $\alpha_i \smile \alpha_j = \beta_i \smile \beta_j = 0$ for $1 \leq i, j \leq g$. We conclude that the cohomology ring of Σ_g is given by desired form. ■

5.4 Homology and Cohomology of Compact Non-Orientable Surfaces

Proposition 5.4.1 Let $h \geq 1$. The homology of N_h is given by

$$H_n(N_h) = \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ \mathbb{Z}^{h-1} \oplus \mathbb{Z}/2\mathbb{Z} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof Similar to the proof in that of Σ_g , cut an open disc along the middle of the CW complex of N_h as follows



and again label the green part U and the orange part V . Then apply Mayer-Vietoris sequence to acquire a similar exact sequence

$$0 \longrightarrow \tilde{H}_2(\Sigma_g) \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^h \longrightarrow \tilde{H}_1(\Sigma_g) \longrightarrow 0$$

together with $\tilde{H}_0(N_h) \cong 0$. Notice that the third non-zero term counting from the left is now $\mathbb{Z}^h$ instead of $\mathbb{Z}^{2g}$ as in the torus because the boundary circle is the wedge sum of h circles labelled $a_1b_1, \dots, a_hb_h$. The map $\mathbb{Z}$ to $\mathbb{Z}^h$ is now given by sending the generator 1 to

$$2(a_1 + b_1 + \dots + a_h + b_h)$$

The Smith Normal form of the matrix is an $h \times 1$ matrix with 2 at the first entry and 0 everywhere else. In particular, it means that this map is injective so that $\tilde{H}_2(N_h) \rightarrow \mathbb{Z}$ is the 0 map so that $\tilde{H}_2(N_h) \cong 0$. Now it remains an exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^h \longrightarrow \tilde{H}_1(N_h) \longrightarrow 0$$

The image of the matrix is $2\mathbb{Z}$ and by exactness this is the kernel of the map $\mathbb{Z}^h \rightarrow \tilde{H}_1(N_h)$. Thus we have an isomorphism

$$\tilde{H}_1(N_h) \cong \mathbb{Z}^{h-1} \oplus \mathbb{Z}/2\mathbb{Z}$$

and so we conclude. ■

Corollary 5.4.2 The surfaces N_h for $h \geq 1$ is non-orientable.

Proof Notice that removing a small closed disk from $\mathbb{RP}^2$ yields a space homeomorphic to the open Mobius strip. It follows that for $h > 0$, the space N_h contains the open Mobius strip as a subspace. Since the Mobius strip is non-orientable, N_h is also non-orientable. ■

Proposition 5.4.3 Let $h \geq 1$. The singular cohomology of non-orientable surface N_h with coefficients in $\mathbb{Z}$ is given by

$$H^n(N_h; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ \mathbb{Z}^{h-1} & \text{if } n = 1 \\ \mathbb{Z}/2\mathbb{Z} & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases}$$

Proof We use the universal coefficient theorem in all dimensions. When $n = 0$, we have that

$$\begin{aligned} H^0(N_h; \mathbb{Z}) &\cong \text{Hom}(H_0(N_h; \mathbb{Z}); \mathbb{Z}) \oplus \text{Ext}(H_{-1}(N_h; \mathbb{Z}), \mathbb{Z}) \\ &\cong \text{Hom}(\mathbb{Z}, \mathbb{Z}) \oplus 0 \\ &\cong \mathbb{Z} \end{aligned}$$

When $n = 1$, we have that

$$\begin{aligned} H^1(N_h; \mathbb{Z}) &\cong \text{Hom}(H_1(N_h), \mathbb{Z}) \oplus \text{Ext}(H_0(N_h), \mathbb{Z}) \\ &\cong \text{Hom}(\mathbb{Z}^{h-1} \oplus \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \oplus \text{Ext}(\mathbb{Z}, \mathbb{Z}) \\ &\cong \mathbb{Z}^{h-1} \oplus 0 \\ &\cong \mathbb{Z}^{h-1} \end{aligned}$$

When $n = 2$, we have that

$$\begin{aligned} H^2(N_h; \mathbb{Z}) &\cong \text{Hom}(H_2(N_h), \mathbb{Z}) \oplus \text{Ext}(H_1(N_h), \mathbb{Z}) \\ &\cong \text{Hom}(\mathbb{Z}, \mathbb{Z}) \oplus \text{Ext}(\mathbb{Z}^{h-1} \oplus \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \\ &\cong 0 \oplus \text{Ext}(\mathbb{Z}^{h-1}, \mathbb{Z}) \oplus \text{Ext}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) \\ &\cong 0 \oplus 0 \oplus \mathbb{Z}/2\mathbb{Z} \\ &\cong \mathbb{Z}/2\mathbb{Z} \end{aligned}$$

When $n \geq 3$, we have that

$$\begin{aligned} H^n(N_h; \mathbb{Z}) &\cong \text{Hom}(H_n(N_h), \mathbb{Z}) \oplus \text{Ext}(H_{n-1}(N_h), \mathbb{Z}) \\ &\cong \text{Hom}(0, \mathbb{Z}) \oplus \text{Ext}(0, \mathbb{Z}) \\ &\cong 0 \end{aligned}$$

and so we conclude. ■

Proposition 5.4.4 Let $h \geq 1$. Then there is a graded ring isomorphism

$$H^*(N_h; \mathbb{Z}) \cong \frac{\mathbb{Z}[\alpha_1, \dots, \alpha_h]}{(\alpha_i^3, \alpha_i \alpha_j, \alpha_i^2 - \alpha_j^2 \mid 1 \leq i \neq j \leq n)}$$

where each α_i has degree 1.

5.5 The Euler Characteristic

Corollary 5.5.1 For $g \geq 0$ and $h > 1$, the Euler characteristic of any compact surface is given by

$$\chi(\Sigma_g) = 2 - 2g \quad \text{and} \quad \chi(N_h) = 2 - h$$

Proof It follows directly by repeated applications of the above corollary. ■

Recall that if $p : \tilde{X} \rightarrow X$ is a d -sheeted covering and X is a finite CW complex, then we have the formula

$$\chi(\tilde{X}) = d \cdot \chi(X)$$

6 Handle Decomposition

6.1

Definition 6.1.1 (Handles)

Let $n \in \mathbb{N}$. Let $k \leq n$. An n -dimensional k -handle is the space

$$D^k \times D^{n-k}$$

Definition 6.1.2 (Handle Attachment)

Let M be a compact topological manifold with boundary. Let $\varphi : S^{k-1} \times D^{n-k} \hookrightarrow \partial M$ be an embedding. Define

$$M_\varphi = \frac{M \amalg (D^k \times D^{n-k})}{\varphi(a, b) \sim (a, b)}$$

Proposition 6.1.3

Let M be a compact topological manifold with boundary. Let $\varphi : S^{k-1} \times D^{n-k} \hookrightarrow \partial M$ be an embedding. Then we have

$$\partial M_\varphi = \frac{(\partial M \setminus \varphi(S^{k-1} \times (D^{n-k})^\circ) \amalg (D^k \times S^{n-k-1}))}{\varphi(a, b) \sim (a, b)}$$

Definition 6.1.4 (Isotopy)

Let M be a compact topological manifold with boundary. Let $\varphi_1, \varphi_2 : S^{k-1} \times D^{n-k} \hookrightarrow \partial M$ be embeddings. An isotopy from φ_1 to φ_2 is a homotopy $H : S^{k-1} \times D^{n-k} \times I \rightarrow \partial M$ from φ_1 to φ_2 such that $H(-, -, t)$ is an embedding for all $t \in I$.

Proposition 6.1.5

Let M be a compact topological manifold with boundary. Let $\varphi_1, \varphi_2 : S^{k-1} \times D^{n-k} \hookrightarrow \partial M$ be embeddings. If φ_1 and φ_2 are homotopic, then there is a homeomorphism

$$M_{\varphi_1} \cong M_{\varphi_2}$$

Definition 6.1.6 (Handle Decomposition)

Let M be a compact topological manifold with boundary. A handle decomposition consists of manifolds $M_{-1} = \emptyset, M_0, \dots, M_{\dim(M)} = M$ such that M_k is obtained from M_{k-1} by attaching k -handles.

Theorem 6.1.7 ((Morse))

Let M be a compact smooth manifold. Then M admits a handle decomposition.

Every compact topological manifold of dimension $n \neq 4$ admits a handle decomposition (requires PL structure).

Proposition 6.1.8

Let M be a compact topological manifold with boundary. Let $\varphi : S^{k-1} \times D^{n-k} \hookrightarrow \partial M$ be an embedding. Then there is a homotopy equivalence

$$M_\varphi \simeq \frac{M \amalg D^k}{\varphi|_{S^{k-1} \times \{0\}}(a) \sim a}$$

Corollary 6.1.9

Let M be a compact topological manifold with boundary. Suppose that M admits a handle decomposition. Then M is homotopy equivalent to a CW complex.

Definition 6.1.10 (Handle Chain Complex)

Let M be a compact topological manifold admitting a handle decomposition. Define the handle chain complex $C_{\bullet}^{\text{hdl}}(M)$ by

7 Higher Dimensional Manifolds

7.1 Manifolds of Dimension 3

Proposition 7.1.1

Let M be a connected topological manifold of dimension 3. Then the homology groups of M is given by

$$H_k(M; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \text{ or } 3 \\ \pi_1(M)^{\text{ab}} & \text{if } k = 1 \\ \text{Hom}_{\mathbb{Z}}(\pi_1(M)^{\text{ab}}, \mathbb{Z}) & \text{if } k = 2 \\ 0 & \text{otherwise} \end{cases}$$

(Uses Algebraic Topology 3)

Proposition 7.1.2

Let M be a connected topological manifold of dimension 3. If M is simply connected, then M is homotopy equivalent to S^3 .

Proof

Since $\pi_1(M) = 0$, we know that M is 1-connected. By Hurewicz's theorem, we have an isomorphism $\pi_2(M) \cong H_2(M) = \text{Hom}_{\mathbb{Z}}(\pi_1(M), \mathbb{Z}) = 0$. Hence M is 2-connected. By Hurewicz's theorem again, we have an isomorphism $\pi_3(M) \cong H_3(M) = \mathbb{Z}$. Let $f : S^3 \rightarrow M$ be a generator of $\pi_3(M)$. Then f induces an isomorphism on all homotopy groups. Since M admits a unique PL structure, it is a CW complex. Hence by Whitehead's theorem, the weak homotopy equivalence $f : S^3 \rightarrow M$ is a homotopy equivalence. ■

<https://pi.math.cornell.edu/hatcher/Papers/3Msurvey.pdf>

7.2 The Intersection Form

Definition 7.2.1 (The Intersection Form)

Let M be a compact orientable topological manifold of dimension $4k$. Define the intersection form of M to be the map

$$Q_M : H^{2k}(M; \mathbb{Z})_{\text{free}} \times H^{2k}(M; \mathbb{Z})_{\text{free}} \rightarrow \mathbb{Z}$$

given by $Q_M(\alpha, \beta) = (\alpha \smile \beta) \frown [M]$

Proposition 7.2.2

Let M be a compact orientable topological manifold of dimension $4k$. Then Q_M is a non-degenerate symmetric bilinear form.

7.3 Manifolds of Dimension 4